MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM, BB-593/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sealed nickel cadmium rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49436.

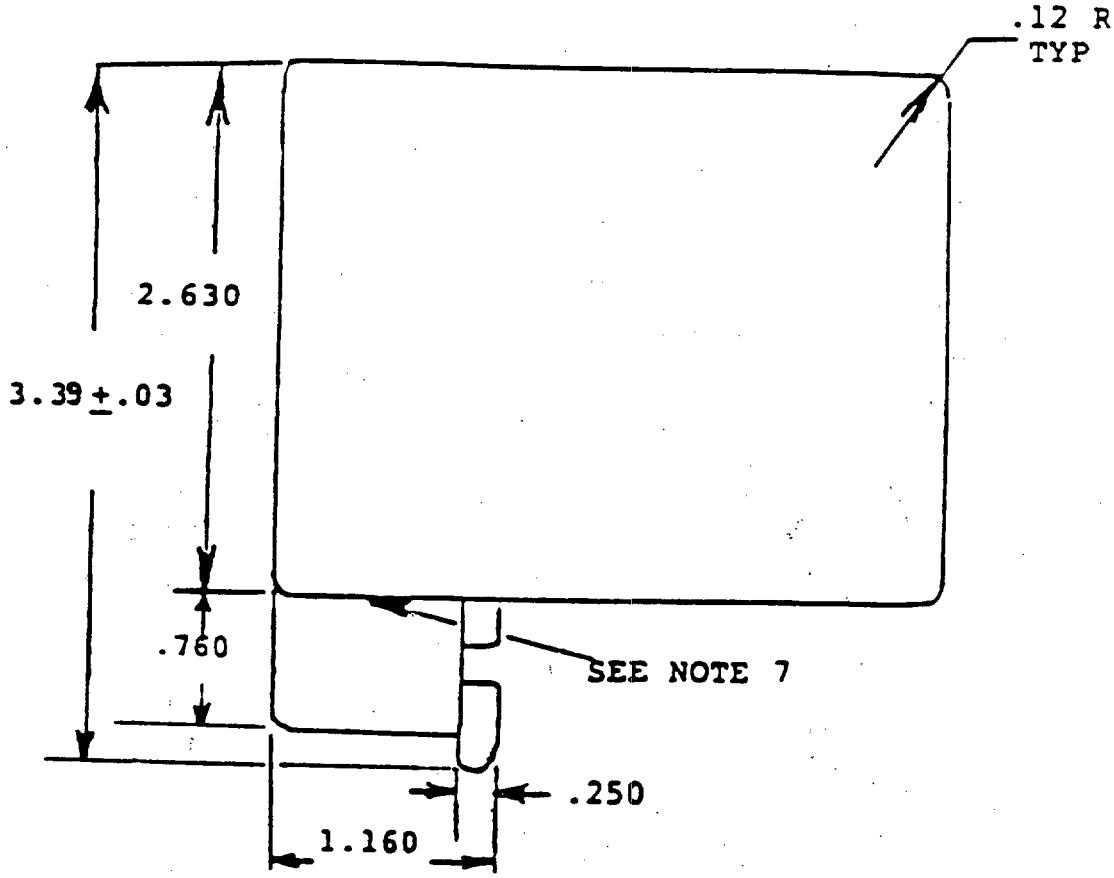


Figure 1. Side view of BB-593/U battery.

AMSC N/A

FSC 6140

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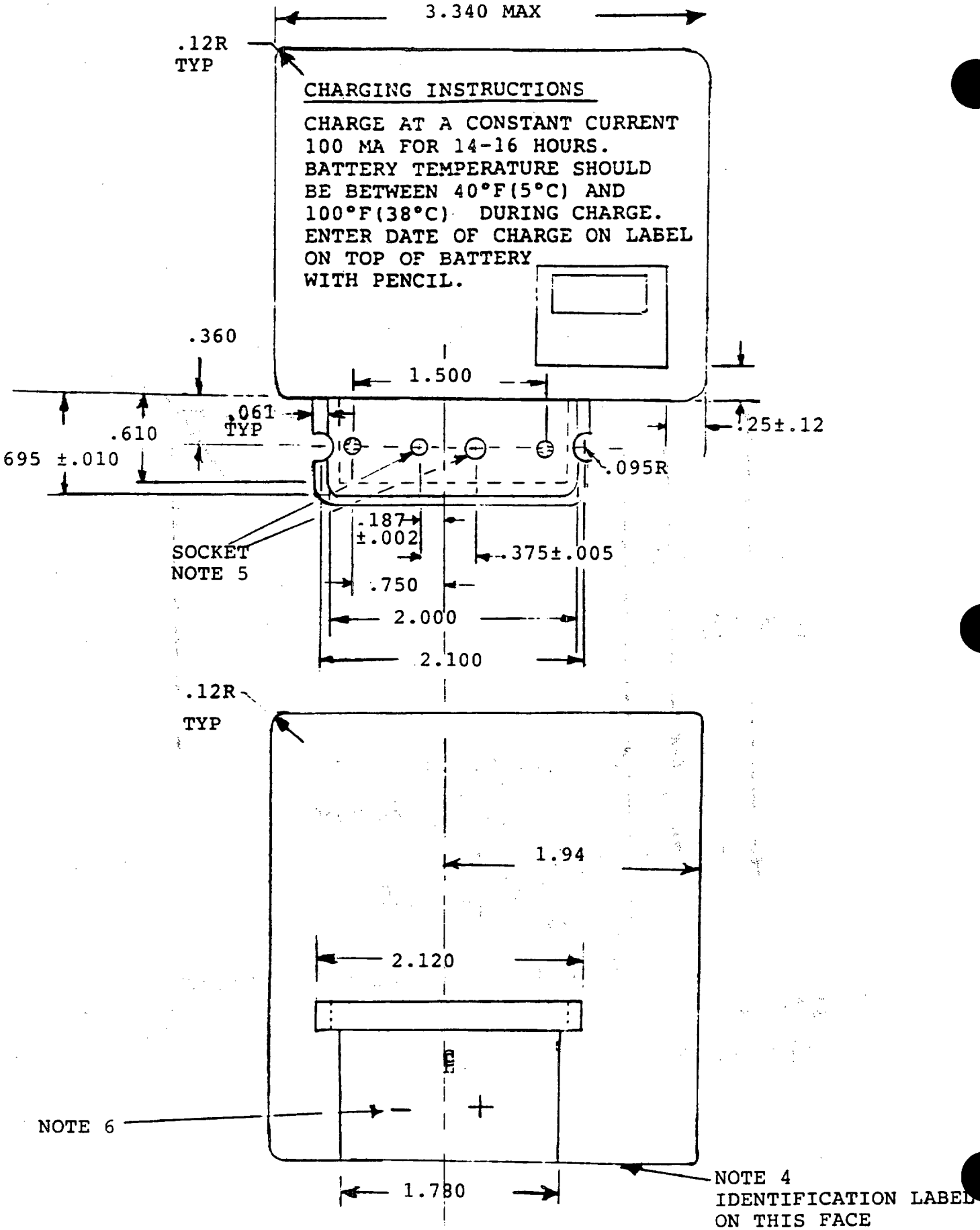


FIGURE 2. TOP AND FRONT VIEW OF BB593/U BATTERY

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DRAWING NOTES:

1. All dimensions in inches.

2. Tolerances on decimals: ± .015 inch, unless otherwise specified.

3. Dimensioning shall be per USAS-114.5 Y14.5 1966.

4. Stencil (or equivalent) identification and charging instructions of Figure 2. Manufacturers must fill in applicable instructions in parenthesis. Marking color shall be white.

5. The battery connector shall be a 2 pin socket connector, type IV per MIL-B-18.

6. The "+" and "-" signs shall be raised 1/16 inch and be 1/4 inch in size.

7. A pressure relief mechanism shall be located in the case, with the preferred location under the connector. It shall release gas before case damage is incurred.

REQUIREMENTS:

The battery connector shall be capable of withstanding a minimum of 500 insertions and withdrawals with its mating connector without evidence of breakage, distortions, or loss of electrical performance. The socket shall be plated to prevent corrosion. The force required to secure full engagement of the stud gage and socket shall not exceed 12 pounds and the force required for disengagement shall not exceed 12 pounds nor be less than 2 pounds when tested during 500 insertions and withdrawals. The contact resistance between the plug gage and socket shall not exceed 0.005 ohm.

To determine conformance to the above requirements, the following tests shall be conducted:

Insertion and withdrawal numbers 1 through 250 shall be made with mating-plug gage "A" having one pin 0.1270 to 0.1273 inch in diameter while the second pin shall be 0.1580 to 0.1583 inches in diameter. Insertion and withdrawal numbers 251 through 500 shall be made with mating-plug gage "B" having one pin 0.1230 to 0.1233 inches in diameter, while the second pin shall be 0.1540 to 0.1543 inches in diameter. Insertion and withdrawal force measurements shall be made during insertion and withdrawal number 1,125,251, 375, and 500. Contact resistance shall be measured after insertions number 1, 250, 251, and 500 while a current of 5 amperes is flowing through the point of contact.

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Mounting screws for connector: Two (2) mounting screws shall be furnished with the battery connector. They shall be corrosion resistant steel, class 303, condition A per ASTM-A484. Mounting screws shall be tightened to prevent free movement of the socket connector.

Rated Capacity (C_5)	1.0 AH
Voltage (nominal):	21.5 volts
Weight (maximum):	0.9 kg (2.0 lb)
Connector:	Two hole-socket terminal per MIL-B-18
Number and type of cells:	18 sealed "3/5 C_S" size cells or equal in series
Charge rate for testing and on battery (C/10 for 14 hrs.)	.10 amps

Terminal markings: Schematic drawing is not applicable.

Electrical leakage: Not applicable.

The identification label shall be located on the bottom face of the battery. The following information shall be on the label:

BATTERY RECHARGEABLE BB-593/U
NICKEL-CADMIUM SEALED
21.6 VOLTS 1.00 AMP-HOURS AT 0.2 AMPS
NSN ( )
(CONTRACT NUMBER)
(DATE CODE) (SERIAL NUMBER)
(MANUFACTURER)
(LOCATION)

Information within the parenthesis shall be filled in by the manufacturer, as applicable.

TESTS:
Cycle life test: Except that the electrical leakage is not applicable.

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Battery case vent test: Fixture shall not be required. The battery shall meet the requirement as described in paragraph 3.4.2 of the basic specification.

Discharge Rates	5 Second Voltage (V)	Final Voltage (V)	Current (Amps)
C/5	-------	18.0	.200
$1C_5$	-------	18.0	1.0

Test	Discharge Rate	Capacity Requirements (AH)
Full Capacity Discharge	$1C_5$	0.85
Cycle Life, 224 cycles	$1C_5$	0.68
Overcharge, 24 hours	C/5	1.0
High Rate Discharge	Not applicable	----
Low Temperature Discharge at $-17^{o}C$ ($0^{o}F$)	C/5	0.8
Retention of Charge 7 days at $51^{o}C$ ($124^{o}F$)	$1C_5$	0.42
Pulse Discharge	Not applicable	----
Vibration	C/5	1.0

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NOTES:

1. Unless otherwise specified, all values are minimum.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
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